

Smartphone-Only Estimation of Elevator Vertical Displacement with Calibrated Confidence Intervals

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Knowing which floor a smartphone user is on remains an open problem on devices that lack a pressure sensor, and the single most common floor-change event — the elevator ride — is exactly where the three workhorses of indoor localisation each break down. Pedestrian dead reckoning and step counting estimate horizontal trajectory; the cabin removes the steps and the accelerometer cannot be integrated vertically without unbounded drift. Wi-Fi fingerprinting demands a per-floor survey, degrades whenever the radio environment drifts, and is shielded by the cabin itself. Barometric altimetry resolves a floor in principle but assumes a pressure sensor that entry-level phones omit, and demands re-referencing against weather drift on phones that carry one.

We present a smartphone-only pipeline that estimates the signed vertical displacement Δh of every elevator ride a user takes and attaches a calibrated 90 % confidence interval to it, using the phone’s accelerometer alone. The pipeline rests on one observation. The mechanical limits of an elevator cabin force the gravity-removed acceleration trace a phone records during a ride into a small parametric family of basis functions. We decompose every candidate ride against this family in two stages — segmentation of hours-long inertial streams into discrete up / down ride intervals, and per-ride prediction of Δh with a per-ride uncertainty. Each interval is distribution-free with finite-sample coverage guarantees, adapts to ride duration, gravity-vector stability, and fit quality, and remains valid under both white sensor noise and the coloured (drift- and vibration-driven) noise that dominates real hand-held recordings. An optional add-on fuses the phone’s orientation sensor on top of the accelerometer pipeline for further accuracy gains.

We evaluate on [TODO: 150] experiments spanning [TODO: TBD] elevators in [TODO: TBD] buildings (residential, office, hotel), with a strict held-out blind test. Median CI half-width on the held-out test stays below [TODO: ± 0.9 m] on short rides at 90 % coverage, an improvement of [TODO: TBD %] over the dataset’s prior best. We open-source the pipeline, dataset, and evaluation harness.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing**; • **Computing methodologies** → *Sensor data processing*.

Additional Key Words and Phrases: smartphone sensing, inertial measurement, elevator, vertical localization, conformal prediction, uncertainty quantification, indoor positioning, building inventory

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1 Introduction

Indoor location services place a smartphone within metres of the right room on the horizontal plane, using Wi-Fi fingerprinting, BLE beacons, and visual pedestrian dead reckoning [8]. The vertical axis lags behind. Knowing the user’s floor matters for emergency dispatch, which routes responders to a floor rather than to a building entrance. It matters for venue-aware advertising and indoor delivery in malls and office towers, and for augmented-reality experiences anchored to a specific floor of a museum or conference centre.

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Three structural limitations make floor estimation harder than its horizontal sibling. (i) Step counting and pedestrian dead reckoning estimate horizontal trajectory and report nothing about height [5]. (ii) Wi-Fi fingerprinting separates floors only where the survey covered every floor and the radio environment has not drifted since [2]. (iii) Barometric altimetry resolves a floor in principle, but the pressure sensor is absent on entry-level phones and demands re-referencing against weather drift on phones that carry one [10]. The single most common floor-change event — the elevator ride — sits in the regime where every horizontal-localisation signal degrades at once.

Elevators are not arbitrary motion. We found that the mechanical limits of the cabin force the gravity-removed acceleration trace a phone records during a ride into a small parametric family of expected functions. Section 2 characterises this family precisely; what matters in the introduction is that the family is small enough to enumerate and structured enough to fit a candidate ride against. Every claim in this paper rests on it.

Two capabilities follow from the family. The first — the focus of this paper — is per-ride. Given an inertial recording of a single elevator ride, return its signed vertical displacement Δh together with a calibrated 90 % confidence interval. The second is building-level. From inertial recordings collected by passersby, classify whether a building has an elevator at all, and, when it does, identify the elevator type by the family parameters its rides span. The combination converts a smartphone into both a per-ride displacement sensor and a passively built building inventory tool; Section 3 develops the building-level capability.

Our pipeline has two stages. Segmentation extracts ride intervals from a continuous, possibly hours-long, inertial stream. Prediction returns Δh for each interval together with a per-ride uncertainty. The pipeline uses the phone’s accelerometer alone — no barometer at inference time, no Wi-Fi, no per-building calibration — and an optional add-on (§4) fuses the phone’s orientation sensor on top of the accelerometer pipeline for further accuracy. The per-ride confidence interval is distribution-free, carries finite-sample coverage guarantees, and remains valid under both white sensor noise and the coloured (drift- and vibration-driven) noise that dominates real hand-held recordings.

We evaluate on [TODO: 150] experiments spanning [TODO: TBD] elevators across [TODO: TBD] buildings (residential, office, hotel), with a strict held-out blind test. The system retains its nominal 90 % coverage on the blind test. The median CI half-width stays below [TODO: ± 0.9 m] on short rides, and per-bin error improves on the dataset’s prior best by [TODO: TBD %]. Section 5 reports the full coverage and error statistics.

Contributions.

- (1) **Per-ride displacement with calibrated coverage.** We deliver 90 %, distribution-free, finite-sample confidence intervals on Δh from accelerometer data alone, valid under both white sensor noise and the coloured noise that dominates hand-held recordings.
- (2) **Family-of-basis-functions framework.** We characterise the family of expected gravity-removed acceleration traces for an elevator ride from the cabin’s mechanical limits, and use the same family as the substrate for both segmentation and per-ride prediction.
- (3) **Passively built building inventory.** We extend the family-fitting machinery to a building-level capability that determines whether a building has an elevator and classifies its type from the family parameters of its rides.
- (4) **Orientation-sensor add-on.** We describe an optional add-on that fuses the phone’s orientation sensor on top of the accelerometer pipeline, with quantified accuracy gain (§4).
- (5) **Dataset and harness.** We release the multi-building dataset ([TODO: 150] experiments, multiple phones per ride, barometer ground-truth pipeline), the full evaluation harness, and the operator UI used to triage rides.

Roadmap. Section 2 characterises the elevator’s mechanical structure, the physical limits it imposes on a ride, and the family of expected acceleration functions those limits force. Section 3 presents the building-level inventory capability. Section 4 presents the orientation-sensor addon. Section 5 describes the data collection, the train/test split, and the results. Sections 6 and 7 discuss limitations and conclude.

2 Background: Elevator Kinematics

This section derives the family of acceleration functions an elevator ride can produce on a smartphone. We start from the cabin’s mechanical parts and the physical constraints they impose (§2.1), translate those constraints into bounds on jerk, acceleration, and velocity (§2.2), and then write down the seven-segment S-curve that the bounds force every ride to follow (§2.3). We close with the closed-form relations between machine parameters and the time constants of the ride (§2.4) and the parametric family that this paper’s segmentation and prediction stages fit (§2.5); the heavy algebra lives in Appendix A.

2.1 How a passenger elevator works

A passenger elevator is, mechanically, a single moving piece: a cabin (the *car*) hanging on a steel rope that runs up over a sheave at the top of the shaft and down the other side to a counterweight (Figure 1). A motor turns the sheave; the cabin and counterweight rise and fall in opposite directions. The counterweight is sized to match the empty car plus roughly half the rated payload, so the motor only has to drive the small mass difference and the inertia of the moving parts. A brake clamps the sheave shaft whenever the motor is unpowered or anything goes wrong. Three machine pieces decide how the cabin can move: the rope (it stretches), the motor (it has a torque ceiling), and the brake (it has a maximum stopping distance fixed by the headroom h_1). The next subsection turns each into a hard limit on the cabin’s motion.

2.2 Physical limitations and what they bound

Four mechanical realities turn into four hard limits on the cabin’s motion. Each is the source of one of the parameters our pipeline fits.

The rope sets the maximum jerk. The main rope stretches. If the motor changes torque too quickly, the cabin starts to bounce on the rope: the passenger feels it as a thump, and the rope itself fatigues over time. Drive controllers therefore ramp torque through a slew-rate limit [9]. Slew-rate-limiting the torque is the same as putting a ceiling on the jerk $\ddot{h}$ the cabin feels. The ride-comfort standard ISO 18738 pins this at $j_{\max} \approx 1.0\text{--}1.6 \text{ m/s}^3$ for passenger elevators [6].

The motor sets the maximum acceleration. At constant velocity the motor only has to fight friction and the small mass mismatch between cabin and counterweight. When it accelerates, it has to push the entire moving mass. The motor has a torque ceiling, so cabin acceleration is also capped: $a_{\max} \approx 0.8\text{--}1.5 \text{ m/s}^2$ on commercial traction elevators. Older geared machines sit at the low end; newer gearless ones at the high end.

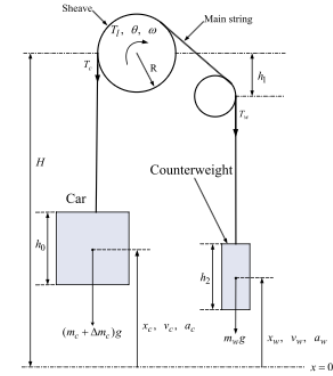


Fig. 1. A traction elevator at a glance. A sheave of radius R couples the car ($m_c + \Delta m_c$) to a counterweight (m_w). The motor delivers torque T_j at angular velocity ω ; rope tensions T_v and T_w load the sheave. H is the hoistway height; h_0 and h_2 are the cabin and counterweight heights; h_1 is the headroom budget above the sheave that the brake must respect.

The brake sets the maximum cruise velocity. If anything goes wrong — a slack rope, an overspeed sensor trip, loss of power — the cabin has to stop before it hits the top of the shaft. The brake decelerates the cabin at some rate a_{brake} and has at most the headroom h_1 above the top floor to do it in. Energy says $v_{\text{max}}^2 \leq 2 a_{\text{brake}} h_1$, which in practice limits cruise velocity to $v_{\text{max}} \approx 1.0\text{--}2.5$ m/s in low- and mid-rise buildings (CIBSE Guide D [4]). Faster cabins exist in skyscrapers, but they require taller headroom and shaft-mounted buffers.

Only the height changes from ride to ride. Once the elevator is built, j_{max} , a_{max} , and v_{max} are fixed for the building — often fixed for the manufacturer and service class. The only thing that changes ride to ride is the floor-to-floor height the cabin has to cover, which we write H (metres, signed for direction). Putting the three limits together, the cabin obeys

$$|\ddot{h}(t)| \leq j_{\text{max}}, \quad |\dot{h}(t)| \leq a_{\text{max}}, \quad |h(t)| \leq v_{\text{max}}, \quad (1)$$

where $\ddot{h} = a$ is the gravity-removed vertical acceleration the phone measures.

2.3 Why the ride is an S-curve

The optimisation problem. The drive solves a simple problem: move the cabin H metres, starting and ending at rest, in the shortest possible time, without ever breaking the three limits in (1). Write the cabin's altitude as $h(t)$, its velocity as $v(t) = \dot{h}(t)$, its acceleration as $a(t) = \ddot{h}(t)$, and the controller's command as the jerk $j(t) = \ddot{a}(t)$. The controller picks $j(t)$; everything else follows by integration. Formally we solve

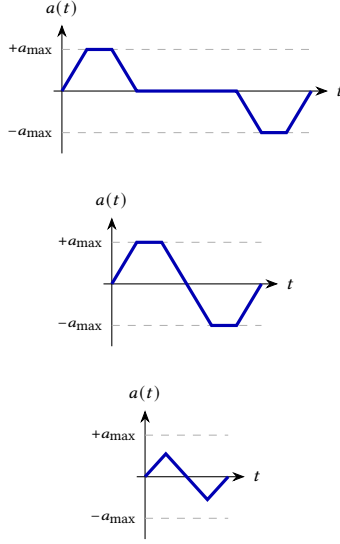
$$\begin{aligned} & \underset{T, j(\cdot)}{\text{minimize}} && T \\ & \text{subject to} && h(0) = 0, \quad h(T) = H, \quad v(0) = v(T) = 0, \\ & && |j(t)| \leq j_{\text{max}}, \quad |a(t)| \leq a_{\text{max}}, \quad |v(t)| \leq v_{\text{max}}, \\ & && \forall t \in [0, T]. \end{aligned} \quad (2)$$

The result. Pontryagin's maximum principle, applied to (2), forces the optimal command to be *bang-bang* on the highest-derivative input. At every instant, the optimal jerk takes only three values,

$$j^*(t) \in \{-j_{\text{max}}, 0, +j_{\text{max}}\}, \quad (3)$$

with $j^* = 0$ holding the cabin against $|a| = a_{\text{max}}$ or $|v| = v_{\text{max}}$ when one of the lower-order limits is itself saturated. Intuitively, any moment spent at an interior value $|j| < j_{\text{max}}$ is wasted: the cabin could have been pushing harder and finished sooner. Appendix B sketches the derivation; we use (3) below.

The S-curve and its three regimes. Stitching the bang-bang jerk command (3) together with the three limits yields the canonical *S-curve* acceleration profile. The floor-to-floor height H alone fixes how many of the limits a ride saturates, and so splits every ride into one of three regimes — long, medium, or short — shown side by side in Figure 2.

**Long ride — 7 segments.**

$$T_v > 0, \quad T_a > 0$$

All three limits saturate; a flat $v_{\max}$ cruise splits the two trapezoidal lobes.

Medium ride — 5 segments.

$$T_v = 0, \quad T_a > 0$$

Jerk and acceleration saturate but the cabin never reaches $v_{\max}$; the two lobes touch at the peak-velocity instant.

Short ride — 3 segments.

$$T_v = T_a = 0$$

Only jerk saturates; the two lobes are triangles that meet, with peak $|a| < a_{\max}$.

Fig. 2. The three S-curve regimes of the time-optimal acceleration profile $a(t)$, all produced by the bang-bang jerk command (3) and selected by the floor-to-floor height H . Each row pairs the profile with the time-constant condition that defines its regime; longer H saturates more limits. Closed forms for T_j , T_a , and T_v follow in §2.4.

2.4 From machine parameters to ride-time constants

The bang-bang structure pins every phase of the ride to a closed-form duration in the machine parameters and the floor-to-floor height H . Three durations describe any ride:

$$T_j = \frac{a_{\max}}{j_{\max}}, \quad (4)$$

$$T_a = \max\left(0, \frac{v_{\max}}{a_{\max}} - T_j\right), \quad (5)$$

$$T_v = \max\left(0, \frac{H}{v_{\max}} - 2T_j - T_a\right), \quad (6)$$

and they sum over the seven phases to the total ride duration

$$T_{\text{ride}} = 4T_j + 2T_a + T_v = \frac{H}{v_{\max}} + \frac{v_{\max}}{a_{\max}} + \frac{a_{\max}}{j_{\max}} \quad (\text{trapezoidal regime}). \quad (7)$$

Figure 3 locates the four durations on a single ride. The jerk ramp T_j and the acceleration plateau T_a are constants of the building. T_j depends only on the machine — about 0.77 s for a typical lift ($a_{\max} = 1.0 \text{ m/s}^2$, $j_{\max} = 1.3 \text{ m/s}^3$) — and T_a vanishes on rides too short to reach $v_{\max}$. Only the cruise T_v scales with ride length, growing as $H/v_{\max}$. Equation (7) is the inversion handle later sections use to recover H jointly with the machine parameters; Appendix A derives all four constants from the bounds in (1).

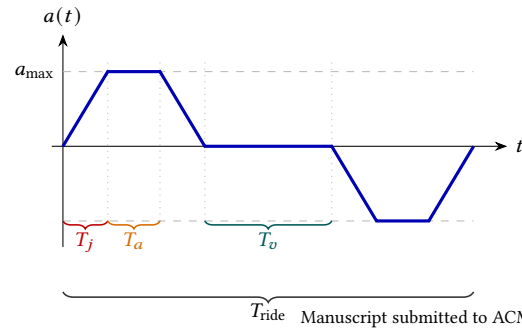


Fig. 3. The four ride-time constants on the acceleration trace $a(t)$ of a trapezoidal ride: jerk ramp T_j , the $a_{\max}$ plateau T_a , the $v_{\max}$ cruise T_v (the flat $a = 0$ gap between lobes), and the total ride T_{ride} .

2.5 The trapezoid basis function and the parametric family

Combining (1)–(7), every ride is fixed by four parameters,

$$\theta = (j_{\max}, a_{\max}, v_{\max}, H), \quad (8)$$

three constants of the building and the ride-specific height H .

The trapezoid basis function. Each acceleration lobe of the S-curve is a trapezoid in time — a ramp up, a flat top, and a ramp down. We normalise it to the trapezoid basis function

$$\tau_{W,f}(t) = \begin{cases} 1 & |t| \leq fW, \\ \frac{W - |t|}{(1-f)W} & fW < |t| \leq W, \\ 0 & |t| > W, \end{cases} \quad (9)$$

of half-width W and flat fraction $f \in [0, 1]$: a unit plateau over $[-fW, fW]$ joined to zero by two linear ramps (Figure 4).

Mapping to the machine parameters. The first lobe spans $2T_j + T_a$, so the machine parameters fix (W, f) exactly:

$$W = T_j + \frac{1}{2}T_a = \frac{a_{\max}}{2j_{\max}} + \frac{1}{2}\max\left(0, \frac{v_{\max}}{a_{\max}} - T_j\right), \quad (10)$$

$$f = \frac{T_a}{T_a + 2T_j} = \frac{\max(0, v_{\max}/a_{\max} - T_j)}{\max(0, v_{\max}/a_{\max} - T_j) + 2T_j}. \quad (11)$$

A short hop that never reaches $v_{\max}$ has $T_a = 0$ and $f = 0$, a pure triangle; a long ride has $T_a \gg T_j$ and $f \rightarrow 1$, a near-rectangular plateau. The two lobes pull apart — their centroid spacing Δt_c exceeds $2W$ — only once a $v_{\max}$ cruise opens between them, the 7-segment regime of §2.3.

The parametric family. Every ride is therefore a pair of opposite-sign trapezoids of shared shape,

$$a_\theta(t) = sA\tau_{W,f}(t - t_{c,1}) - sA\tau_{W,f}(t - t_{c,2}), \quad (12)$$

with amplitude $A = a_{\max}$, direction $s \in \{+1, -1\}$, and centroid spacing $\Delta t_c = t_{c,2} - t_{c,1} = H/v_{\max}$ (Appendix A).

Three facts about this family $\mathcal{F}_{\text{ride}}$ drive the rest of the paper.

(i) Three of its parameters are constants of the building, so inverting enough rides identifies the building (§3). (ii) The two lobes carry equal and opposite area, so $\int_0^{T_{\text{ride}}} a_\theta(t) dt = 0$ — the zero-mean anchor for any ZUPT-style predictor. (iii) The lobe spacing $\Delta t_c = H/v_{\max}$ is a scale-free estimator of H , so the matched-filter detector needs no per-building amplitude calibration.

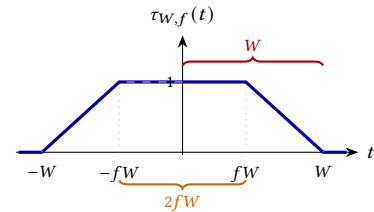
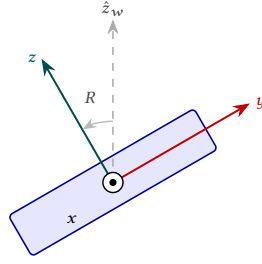


Fig. 4. The trapezoid basis function $\tau_{W,f}(t)$ of (9).

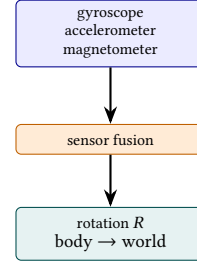
2.6 Smartphone sensors

The pipeline reads two phone sensors (Figure 5): the accelerometer, which it always uses, and the orientation sensor, which feeds the optional add-on of §4.

The accelerometer. Every phone carries a six-axis MEMS IMU [3]. Its TYPE_LINEAR_ACCELERATION stream reports the gravity-removed acceleration as a three-vector $\mathbf{a} = (a_x, a_y, a_z)$ along the phone's *body axes* — the x, y, z fixed to



(a) The accelerometer measures along the phone's body axes x, y, z , tilted by an unknown rotation R from the world vertical $\hat{z}_w$.



(b) The orientation sensor fuses the gyroscope, accelerometer, and magnetometer into the body-to-world rotation R .

Fig. 5. The two smartphone sensors the pipeline reads: the accelerometer's body-frame signal (a) and the orientation sensor's fused rotation (b).

the case (Figure 5a). The sensor-hub rate varies by device, so we resample every recording to a uniform 100 Hz grid ($\Delta t = 10$ ms). The noise floor is white, but a hand-held phone adds coloured drift and 1–50 Hz vibration, so later sections fit the noise model on real ride residuals.

The unknown vertical. An elevator ride is purely vertical: the cabin accelerates along the world up-axis $\hat{z}_w$, and that one scalar $a(t)$ is all the pipeline wants. The accelerometer, though, measures along the phone's body axes, and a hand-held phone tilts arbitrarily. An unknown rotation R separates the two frames, so the vertical signal spreads across all three channels, $\mathbf{a} = R^T \hat{z}_w$ — the body channel a_z is not the world vertical. Recovering $a(t)$ means recovering which body direction points up; the accelerometer-only pipeline reads it from gravity in the stationary windows around each ride.

The orientation sensor. The orientation sensor recovers that direction directly by fusing three sensors (Figure 5b). The gyroscope reports angular rate, which integrates into smooth rotation but drifts over seconds. The accelerometer's gravity reading pins the two tilt axes and the magnetometer pins heading, so the fusion cancels that drift. Its output, at every sample, is a quaternion — the rotation R from the body frame to the world frame. With R , the pipeline rotates each acceleration sample into the world frame and reads the vertical directly, $a = \hat{z}_w^T R \mathbf{a}$. The add-on of §4 builds on this rotation and quantifies the accuracy it adds.

In short, the pipeline works from a 100 Hz body-frame acceleration stream, optionally paired with a per-sample body-to-world rotation.

3 Building-Level Inventory

4 Orientation-Sensor Addon

5 Evaluation

We evaluate the pipeline on a controlled multi-building field campaign with barometer-derived ground truth.

5.1 Dataset and ground truth

The dataset is a field campaign of [TODO: ≈ 150] elevator rides recorded across [TODO: 7] elevators in [TODO: 6] buildings in central Israel (Table 1). The buildings span residential, office, and hotel cabins, each with a different cabin

geometry, motor, and floor-to-floor height, so no estimator can overfit a single machine. Four phones, carried together, recorded every ride, so one physical event produced four heterogeneous accelerometer traces — different MEMS parts, sampling jitter, and noise floors. At least one of the four phones carries a barometric pressure sensor, which serves as the ground-truth device even when the phone under test has none.

Every ride’s signed displacement Δh comes from the barometer rather than from hand-labelling. We invert the pressure stream to altitude through the ISA barometric formula; a consumer barometer resolves altitude to roughly 10 cm, an order of magnitude finer than the ~ 3 m inter-floor pitch the pipeline must separate. Low-pass filtering and differencing the same altitude trace yields the up, down, and outside interval labels that cut each session into discrete rides. A phone without a barometer borrows the pressure stream of a co-carried device, aligned to it by accelerometer cross-correlation (Appendix C).

Raw barometric altitude drifts with weather and building airflow, so we snap it to the building itself. Each building contributes a *gramushka* table — the floor elevations read from its accordion-folded architectural drawing (Figure 6). We anchor every session at its known start floor and integrate the barometer altitude across each detected ride. Snapping the ride endpoint to the nearest table floor makes the ground-truth Δh a true floor-to-floor displacement. The pipeline flags any snap whose distance exceeds [TODO: TBD] m for manual review rather than forcing it, and the one elevator without a drawing falls back to the drift-corrected barometer Δh (Appendix C).

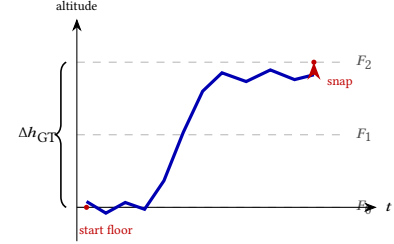


Fig. 6. Snapping the drifting barometer altitude (blue) to the nearest *gramushka* floor line (dashed) yields a drift-free ground-truth Δh .

6 Discussion

7 Conclusion

References

- [1] Michael Athans and Peter L. Falb. 1966. *Optimal Control: An Introduction to the Theory and Its Applications*. McGraw-Hill.
- [2] Paramvir Bahl and Venkata N. Padmanabhan. 2000. RADAR: An In-Building RF-Based User Location and Tracking System. In *Proceedings of IEEE INFOCOM*. 775–784. doi:10.1109/INFCOM.2000.832252
- [3] Stephen Beeby, Graham Ensell, Michael Kraft, and Neil White. 2004. *MEMS Mechanical Sensors*. Artech House.
- [4] Chartered Institution of Building Services Engineers. 2020. *CIBSE Guide D: Transportation Systems in Buildings* (fifth ed.). CIBSE.
- [5] Robert Harle. 2013. A Survey of Indoor Inertial Positioning Systems for Pedestrians. *IEEE Communications Surveys & Tutorials* 15, 3 (2013), 1281–1293. doi:10.1109/SURV.2012.121912.00075
- [6] International Organization for Standardization. 2012. ISO 18738-1:2012 — Measurement of Ride Quality, Part 1: Lifts (Elevators). Standard, ISO.
- [7] Donald E. Kirk. 2004. *Optimal Control Theory: An Introduction* (reprint ed.). Dover Publications.
- [8] Hui Liu, Houshang Darabi, Pat Banerjee, and Jing Liu. 2007. Survey of Wireless Indoor Positioning Techniques and Systems. *IEEE Transactions on Systems, Man, and Cybernetics, Part C* 37, 6 (2007), 1067–1080. doi:10.1109/TSMCC.2007.905750
- [9] Sonja Macfarlane and Elizabeth A. Croft. 2003. Jerk-Bounded Manipulator Trajectory Planning: Design for Real-Time Applications. *IEEE Transactions on Robotics and Automation* 19, 1 (2003), 42–52. doi:10.1109/TRA.2002.807548
- [10] Kartik Muralidharan, Azeem Javed Khan, Archan Misra, Rajesh Krishna Balan, and Sharad Agarwal. 2014. Barometric Phone Sensors: More Hype Than Hope!. In *Proceedings of the 15th Workshop on Mobile Computing Systems and Applications (HotMobile)*. 12:1–12:6. doi:10.1145/2565585.2565596

A Derivations referenced in Section 2.4

This appendix derives the closed forms for T_j , T_a , T_v , the acceleration-phase displacement d_{accel} , and the trapezoidal-regime ride duration (7) from the bounds in (1). Throughout, we assume the trapezoidal regime $T_v > 0$; the triangular

and joined-pulse regimes ($T_v = 0$, with T_a also vanishing in the triangular limit) are the limits $H \downarrow a_{\max} T_j (T_j + T_a)$ of these formulas.

Phase-1 velocity and displacement. On phase 1, $j(t) = +j_{\max}$ with $a(0) = v(0) = 0$. Integrating once, $a(t) = j_{\max} t$, so $a(T_j) = a_{\max}$ forces $T_j = a_{\max} / j_{\max}$, which is (4). Integrating once more, $v(T_j) = \frac{1}{2} j_{\max} T_j^2 = \frac{1}{2} a_{\max} T_j$, and integrating a third time, $h(T_j) = \frac{1}{6} j_{\max} T_j^3 = \frac{1}{6} a_{\max} T_j^2$.

Phase-2 velocity and displacement. On phase 2, $a(t) = a_{\max}$ for $t \in [T_j, T_j + T_a]$. Hence $v(T_j + T_a) = v(T_j) + a_{\max} T_a = a_{\max} (\frac{1}{2} T_j + T_a)$. The displacement added in phase 2 is $\Delta h_2 = v(T_j) T_a + \frac{1}{2} a_{\max} T_a^2$.

Phase-3 velocity and displacement. On phase 3, $j(t) = -j_{\max}$ for $t \in [T_j + T_a, 2T_j + T_a]$, with $a(T_j + T_a) = a_{\max}$. Integrating, $a(t) = a_{\max} - j_{\max}(t - T_j - T_a)$, so $a(2T_j + T_a) = 0$ as required. At the end of phase 3 the velocity is $v(2T_j + T_a) = v(T_j + T_a) + \frac{1}{2} a_{\max} T_j = a_{\max} (T_j + T_a)$, which must equal $v_{\max}$ at the start of cruise. Solving $a_{\max} (T_j + T_a) = v_{\max}$ for T_a gives (5). The displacement added in phase 3 is $\Delta h_3 = v(T_j + T_a) T_j + \frac{1}{2} a_{\max} T_j^2 - \frac{1}{6} j_{\max} T_j^3$.

Acceleration-phase displacement d_{accel} . Summing phases 1–3,

$$d_{\text{accel}} = h(T_j) + \Delta h_2 + \Delta h_3 = \frac{1}{2} v_{\max} (2T_j + T_a),$$

where the simplification uses $j_{\max} T_j = a_{\max}$ and $a_{\max} (T_j + T_a) = v_{\max}$. By symmetry, phases 5–7 traverse the same displacement d_{accel} .

Cruise duration T_v and total ride duration T_{ride} . The cruise phase 4 must cover the remaining $H - 2d_{\text{accel}} = H - v_{\max} (2T_j + T_a)$ at velocity $v_{\max}$:

$$T_v = \frac{H - v_{\max} (2T_j + T_a)}{v_{\max}} = \frac{H}{v_{\max}} - 2T_j - T_a,$$

which is (6). The total ride duration is $T_{\text{ride}} = 4T_j + 2T_a + T_v$; substituting (5) and (6) and simplifying yields the additive form (7).

Centroid spacing of the two acceleration lobes. The first lobe spans $[0, 2T_j + T_a]$ with centroid at $t_{c,1} = T_j + T_a/2$ (by symmetry of the trapezoid). The second lobe spans $[2T_j + T_a + T_v, 4T_j + 2T_a + T_v]$ with centroid at $t_{c,2} = 3T_j + 3T_a/2 + T_v$. The centroid spacing is therefore

$$\Delta t_c = t_{c,2} - t_{c,1} = 2T_j + T_a + T_v = \frac{H}{v_{\max}} - T_a + T_a = \frac{H}{v_{\max}},$$

where the third equality uses (6). This proves the amplitude-free centroid identity used in the basis-function mapping of §2.5: the temporal gap between the two acceleration lobes equals $H/v_{\max}$ and is independent of $j_{\max}$ and $a_{\max}$.

B Time-optimal jerk is bang-bang

This appendix sketches why the optimal-time solution (2) commands a jerk $j^*(t) \in \{-j_{\max}, 0, +j_{\max}\}$. We follow the standard Pontryagin-maximum-principle argument for time-optimal control of a chain of integrators with bounded input [1, 7]; this paper does not depend on the algebra below, only on the result (3).

State and Hamiltonian. Stack the cabin's state as $x = (h, v, a)^\top$ with dynamics $\dot{x} = (v, a, j)^\top$, control $j(t)$, and free terminal time T . The cost is $\int_0^T 1 dt$. Adjoining co-state multipliers $\lambda = (\lambda_h, \lambda_v, \lambda_a)^\top$ gives the Hamiltonian

$$\mathcal{H}(x, \lambda, j) = 1 + \lambda_h v + \lambda_v a + \lambda_a j.$$

The PMP requires $j^*(t)$ to minimise $\mathcal{H}$ over the feasible set $|j| \leq j_{\max}$ at every instant. $\mathcal{H}$ is linear in j , so the minimum is at the boundary:

$$j^*(t) = \begin{cases} +j_{\max}, & \lambda_a(t) < 0, \\ -j_{\max}, & \lambda_a(t) > 0. \end{cases} \quad (13)$$

The co-state equations $\dot{\lambda} = -\partial\mathcal{H}/\partial x$ give $\dot{\lambda}_h = 0$, $\dot{\lambda}_v = -\lambda_h$, $\dot{\lambda}_a = -\lambda_v$, so $\lambda_a(t) = \frac{1}{2}\lambda_h t^2 - \lambda_v(0)t + \lambda_a(0)$ is at most quadratic in t and crosses zero at most twice. Hence j^* has at most two switching times on any unconstrained arc.

State-constraint arcs. When the state-inequality $|a| \leq a_{\max}$ becomes active, the solution slides along the boundary $a \equiv \pm a_{\max}$. On such an arc, the controller must hold $\dot{a} = j = 0$; otherwise the cabin would push past $a_{\max}$. Identical reasoning applies to the velocity constraint $|v| \leq v_{\max}$: when active, $\dot{v} = a = 0$, which in turn forces $j = 0$. Stitching the unconstrained bang-bang arcs together with the constrained-zero arcs gives the three-valued result (3).

Singular-arc ruling. A singular arc would require $\lambda_a(t) \equiv 0$ on a set of positive measure. Differentiating $\lambda_a \equiv 0$ in time gives $\dot{\lambda}_a = -\lambda_v \equiv 0$ and a further differentiation gives $\lambda_h \equiv 0$. But the PMP transversality condition for free terminal time is $\mathcal{H}(T) = 0$, which on a singular arc reduces to $1 = 0$ — a contradiction. No singular arcs exist, so the bang-bang structure is exhaustive.

Resulting profile. With $j^* \in \{-j_{\max}, 0, +j_{\max}\}$ everywhere, the longest profile that includes both signs of the $|a| = a_{\max}$ plateau and the $|v| = v_{\max}$ plateau is the seven-segment S-curve. Suppressing one plateau yields the five-segment regime; suppressing both yields the three-segment regime. Which regime applies depends on the boundary condition $h(T) = H$, as enumerated in §2.3.

C Dataset collection and ground-truth pipeline

This appendix details the field campaign and the ground-truth construction summarised in §5.1.

Buildings and devices. Table 1 lists the [TODO: 7] elevators of the campaign, concentrated in Ra'anana with one building each in Herzeliya and Ramat Gan. The Millenium hotel contributes two physically distinct elevators — an internal cabin and an external one attached to the hotel body — which we treat as separate elevators. Four phones recorded every ride, spanning a mix of Galaxy S23, Pixel, Redmi Note 12, Galaxy A23, and Galaxy Z Flip6 hardware; the experimenters carried all four in one hand or bag, so every device saw near-identical motion, vibration, and acoustics. Figure 7 shows the resulting spread: the elevators range from a 7-floor external hotel cabin to a 23-floor residential tower, with floor spacing that varies within and across buildings.

Logging. An Android logger subscribes to every inertial and environmental sensor the phone exposes and writes one tab-separated row per sample. Table 2 gives the channel schema. The barometer channel (PRS) is the ground-truth signal; we pair any phone without a barometer with a co-carried forBarometer device that has one.

Barometer interval detector. The detector converts pressure to altitude through the ISA formula, low-pass filters it, and differences it into vertical velocity. It marks samples above 0.15 m/s as moving, merges same-sign runs across short pauses, and drops runs too brief or too shallow to be a floor change. It pads the survivors into up and down intervals and fills the gaps between them with outside. The procedure is deterministic, so repeated runs label a recording identically.

Cross-device time synchronisation. Android timestamps a sensor event against device boot time, not wall-clock, so two phones disagree on when one physical event occurred by their boot-time difference — often tens of thousands of

Elevator (internal name)	Type	Location
milleniumHotel	Hotel	Ra'anana
milleniumOutside	Hotel	Ra'anana
acroBuilding	Office	Ra'anana
beitMansour1	Residential	Ra'anana
beitYitzchakiRaanaana	Residential	Ra'anana
BarIlan2Herzeliya	Residential	Herzeliya
Haari3	Residential	Ramat Gan

Table 1. The [TODO: 7] elevators across [TODO: 6] buildings of the field campaign. milleniumHotel and milleniumOutside are two elevators in one hotel; milleniumOutside has no architectural drawing and uses drift-corrected barometer ground truth. beitYitzchakiRaanaana is the held-out blind-test building.

Tag	Sensor	Columns
ACC	Linear accelerometer	x, y, z
GYR	Gyroscope	x, y, z
MAG	Magnetometer	x, y, z
ORI	Rotation-vector quaternion	w, x, y, z
PRS	Barometer	pressure
GPS	GPS fix	lat, lon, alt

Table 2. Sensor channels emitted by the logger. The pipeline reads ACC throughout and ORI for the orientation add-on of §4; PRS supplies ground truth only.

seconds. Aligning the first sample of each log fixes the offset to within the experimenter’s tap latency; a normalised cross-correlation of the two accelerometer-magnitude traces over a ± 10 s window removes the residual. The pipeline bakes the recovered offset into the secondary barometer stream before merging it, so every downstream stage sees one phone with a valid barometer. Figure 8 overlays the two phones’ accelerometer traces for one session after the correction.

Gramushka floor-snapping. Each building contributes a *gramushka* table — “gramushka” names the accordion-folded architectural drawing (Figure 9) — listing every floor and its surveyed elevation. The pipeline reads the session’s start floor from the recording metadata, integrates the temperature-aware barometer altitude across each detected ride, and snaps the ride endpoint to the nearest floor in the table; the ground-truth Δh is the resulting floor-to-floor displacement. The pipeline writes any snap whose distance exceeds [TODO: TBD] m to a per-experiment flag file for manual review rather than applying it silently. The external Millenium elevator has no drawing, so its rides keep the drift-corrected barometer Δh .

Manual audit. A barometer detector occasionally splits one ride in two, merges two into one, or mislabels an HVAC pressure ripple as motion. Every ground-truth file therefore passes at least once through an editor (Figure 10) that overlays the interval labels on altitude, vertical velocity, and the three motion magnitudes, so a human checks each label against several modalities at once.

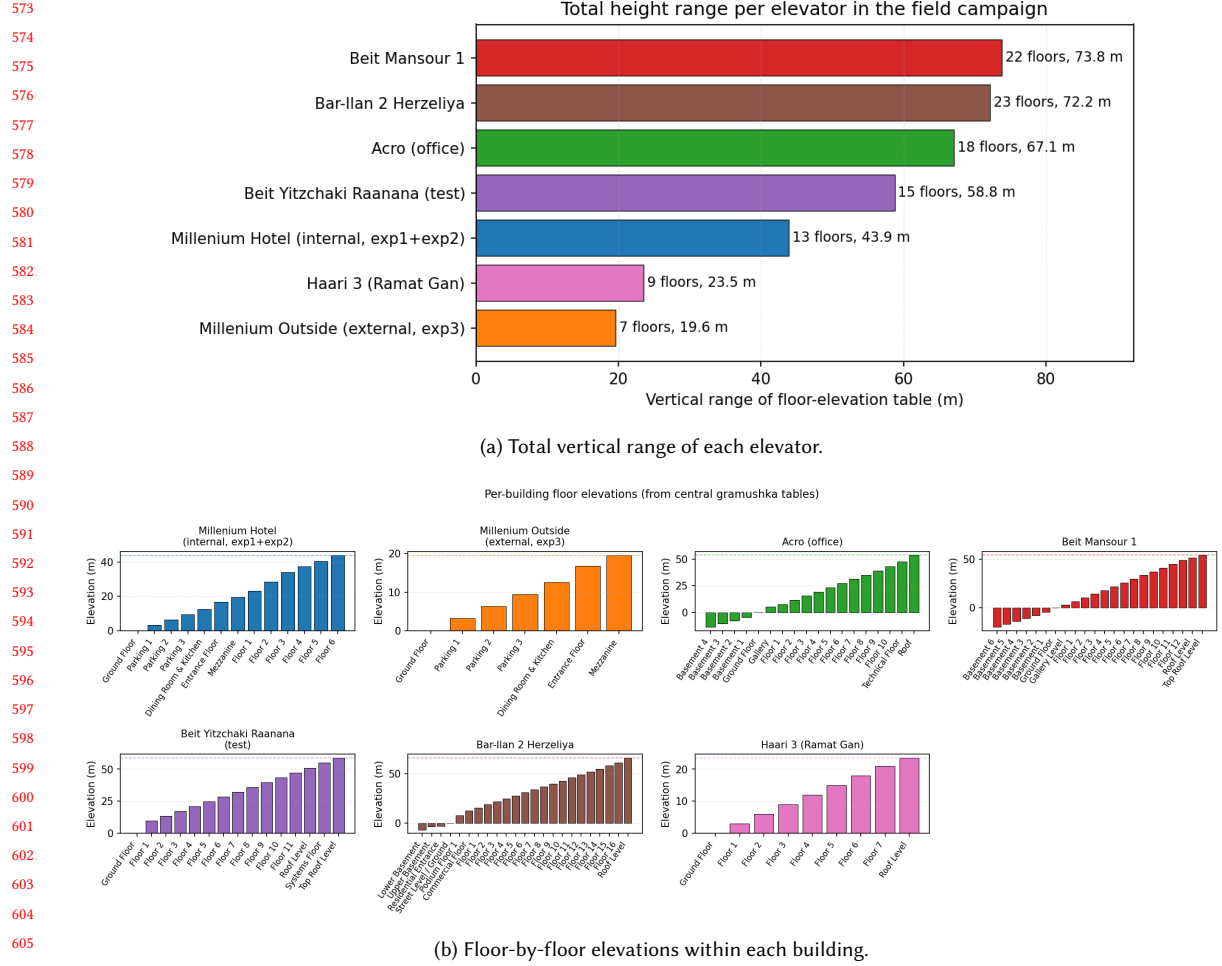


Fig. 7. The campaign's buildings, read from their *gramushka* floor-elevation tables. (a) Elevators range from a 7-floor external hotel cabin to a 23-floor residential tower. (b) Floor spacing varies within and across buildings — dense low-rise residential floors against a sparse high-range office stack.

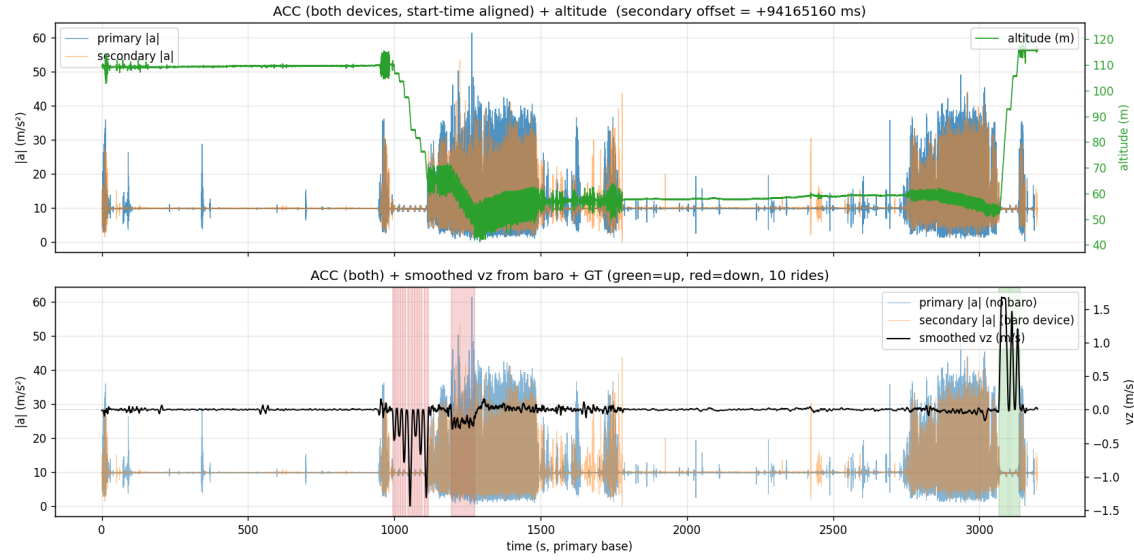


Fig. 8. One recording session after cross-device alignment. **Top:** the primary phone's accelerometer magnitude $|a|$ (blue) and the time-shifted secondary phone's $|a|$ (orange), with the secondary barometer altitude (green); the two traces coincide once the boot-time offset is removed. **Bottom:** the same overlay with the barometer-detected ride intervals shaded (green = up, red = down).

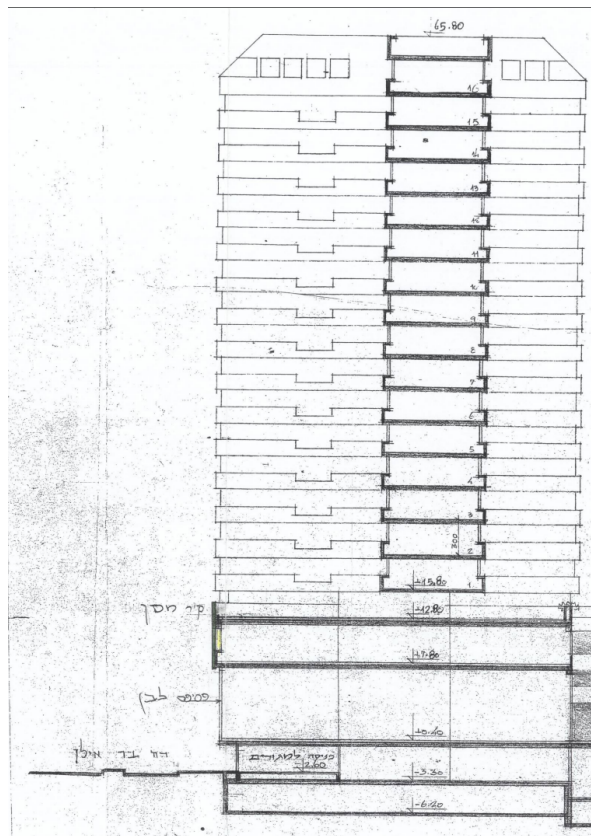


Fig. 9. A *gramushka*: the accordion-folded architectural section drawing of a campaign building (Bar-Ilan 2, Herzeliya). Every floor line carries a surveyed elevation; transcribing those elevations yields the building's floor table.



Fig. 10. The ground-truth editor. Stacked panels show altitude, vertical velocity, and the accelerometer, gyroscope, and magnetometer magnitudes on a shared time axis; ground-truth intervals are overlaid as coloured spans (green = up, red = down, grey = outside). The side table lists every interval for direct editing.